



The Anatomy of Poverty Analysis

Measurement, Determinants, and Policy in Applied Welfare Economics

Course Roadmap

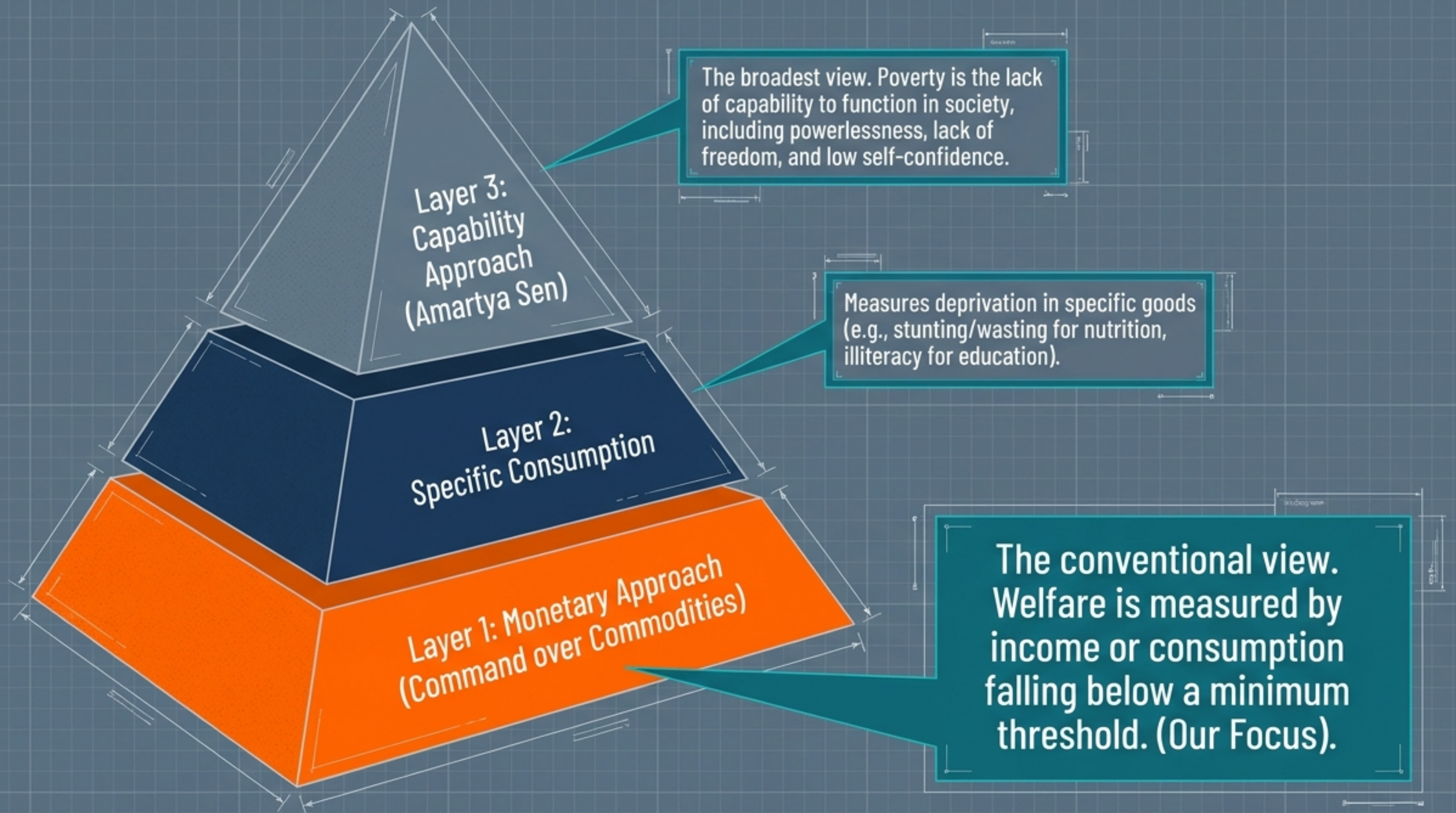
1. Conceptualization
(Defining Well-Being)

2. Operationalization
(Lines & Indices)

3. Distribution
(Inequality &
Robustness)

4. Causation
(Econometric
Determinants)

5. Action
(Pro-Poor Policy)



Monetary Welfare Indicators

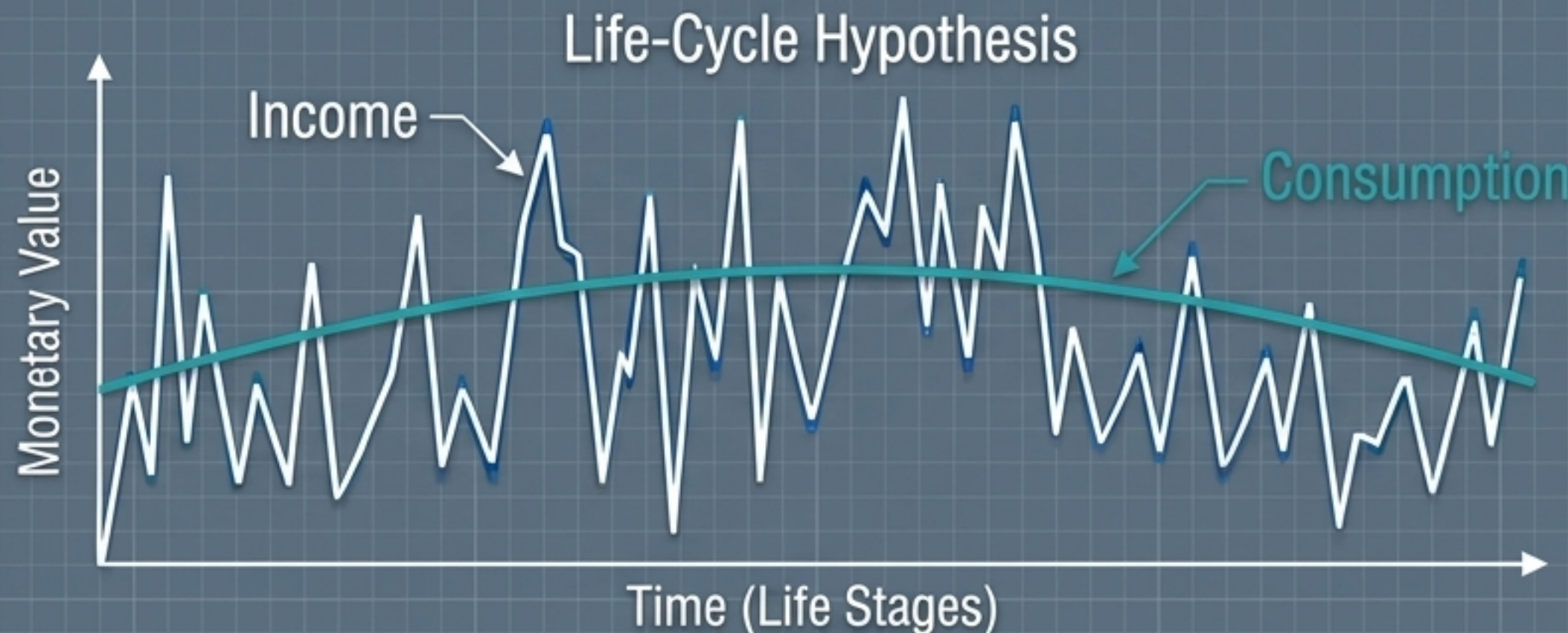
Column 1	Column 2
Income	Consumption (Preferred)
• Easier to measure in formal sectors. • Fails to capture farm income, seasonal volatility, and informal work.	• Better reflects “permanent income” (utility). • Harder to measure due to valuation of durable goods, housing, and own-farm production

Class Work

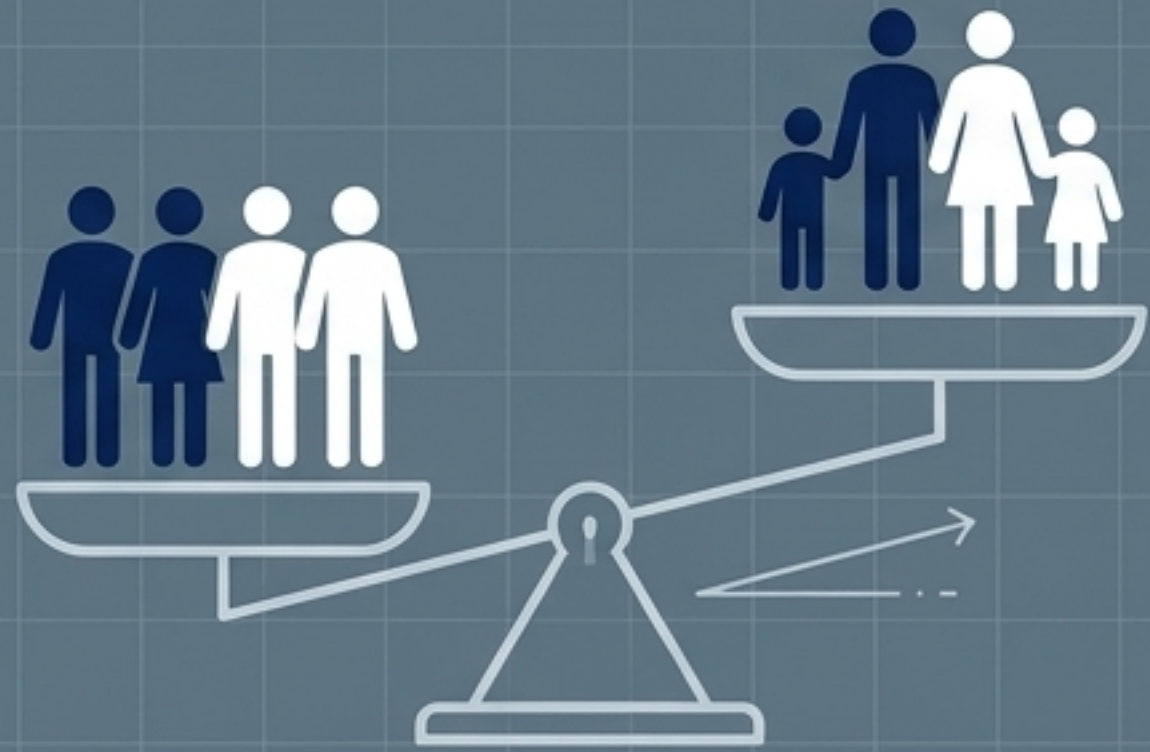
Quiz: A household buys a TV for \$100 this year; it lasts 10 years. How is this recorded in our consumption survey?



Answer: Use the implicit rental value/depreciation, not the purchase price.



Demographic Adjustments: The OECD Equivalence Scale



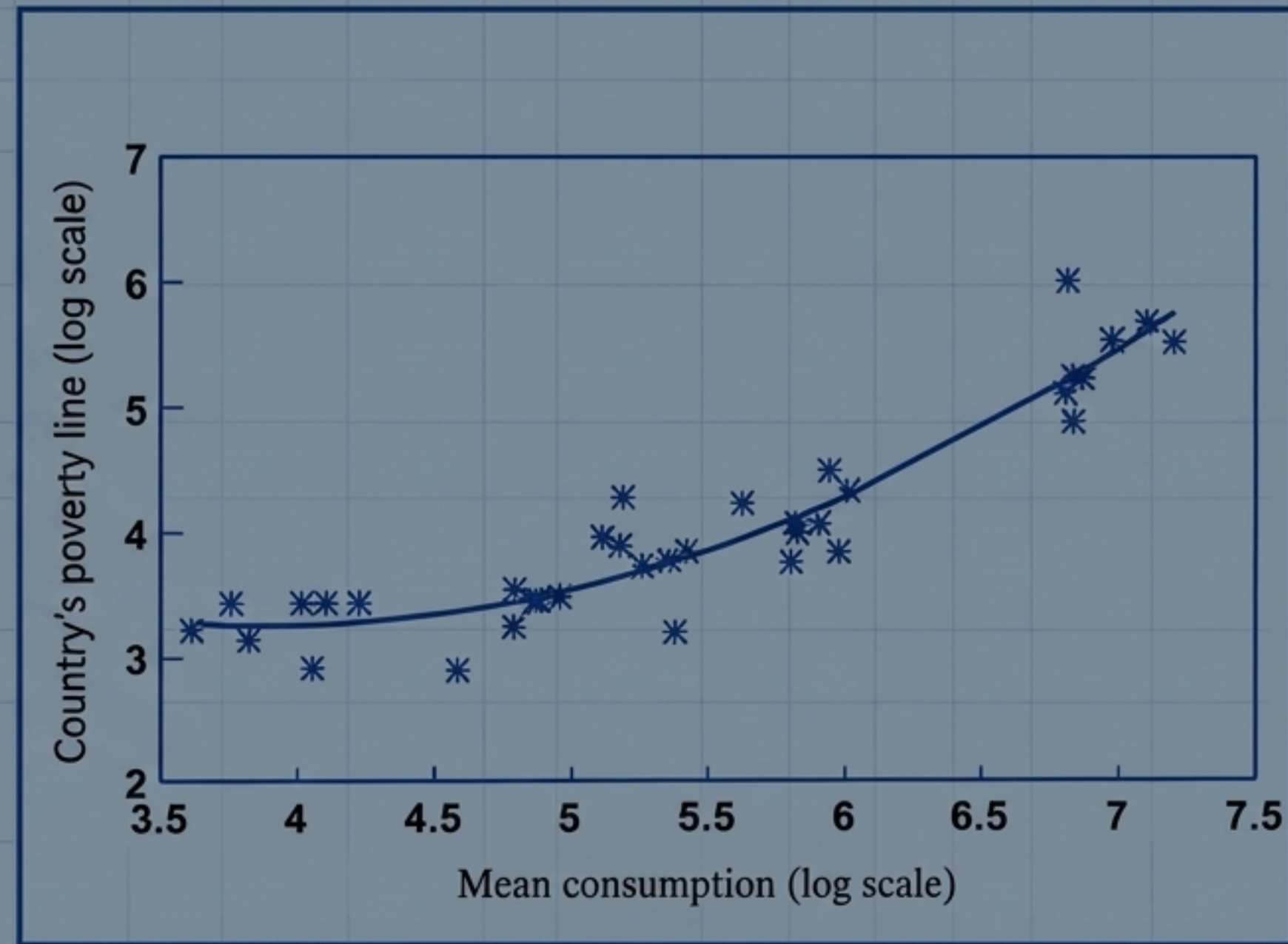
The Problem: Different members have different needs (children vs. adults) and households benefit from economies of scale (shared housing, utilities).

$$AE = 1 + 0.7(N_{\text{adults}} - 1) + 0.5(N_{\text{children}})$$

Adult Equivalents Number of Additional Adults (beyond the first) Number of Children

Impact: A household of 2 adults + 2 children = 2.7 Adult Equivalents, not 4. Using per capita vs. per adult equivalent fundamentally shifts the poverty distribution.

Poverty Line Type	Description
Absolute Poverty Line	Fixed standard of living. Essential for cross-country comparisons (e.g., World Bank's \$1.08/day in 1993 PPP).
Relative Poverty Line	Rises with average income (e.g., 50% of median income).



The Empirical Reality: As countries become richer, they adjust their real poverty lines upward.

Class Question: True or False? The World Bank's \$1/day line is a relative poverty line.

Answer: False. It is absolute, anchored to purchasing power parity.

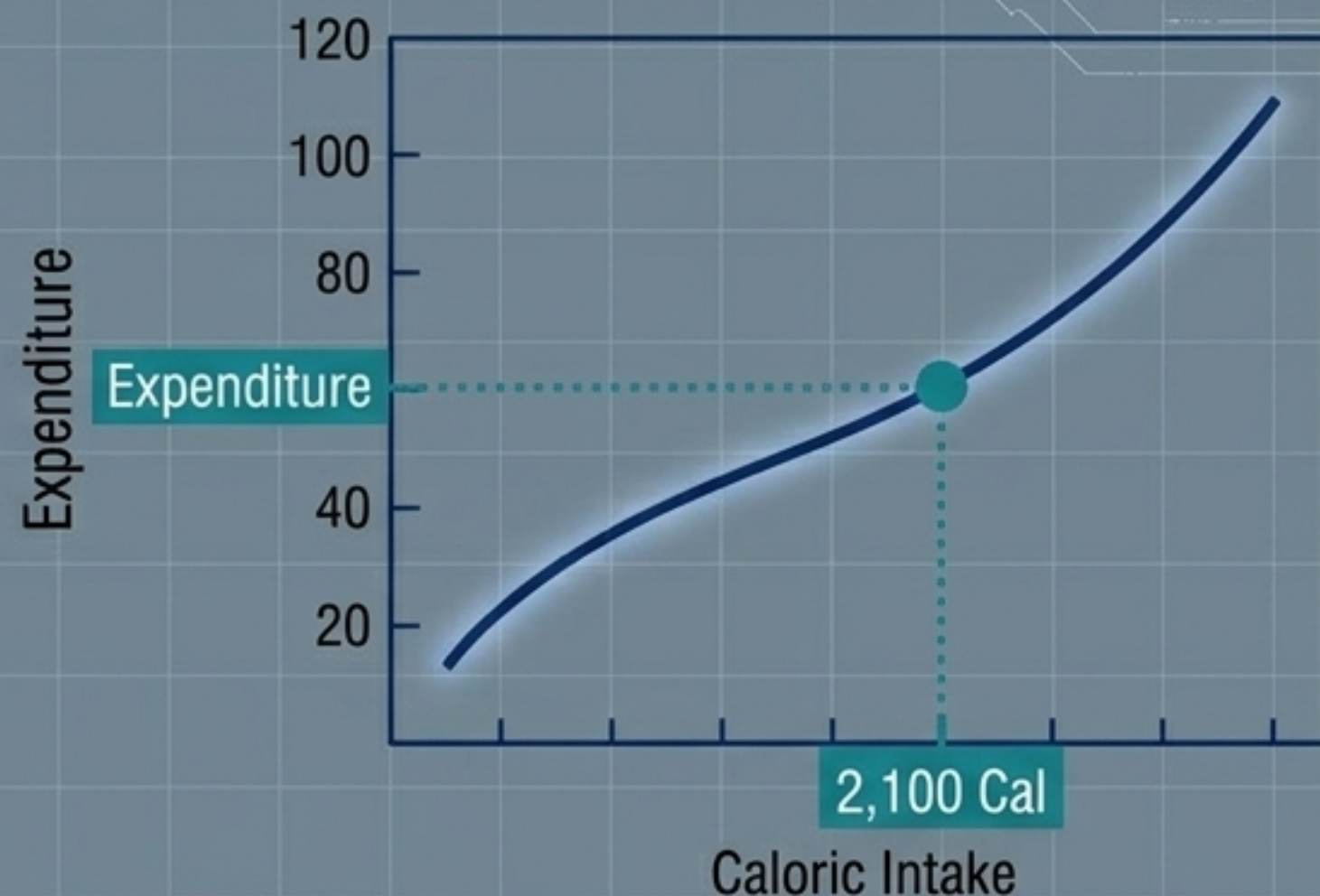
Method 1: Cost of Basic Needs (CBN)

Step 1: Stipulate adequate food bundle (typically 2,100 Calories/person/day).

Step 2: Estimate local cost of this bundle.

Step 3: Add an allowance for essential non-food goods (shelter, clothing).

Method 2: Food Energy Intake (FEI)



The Flaw: Vulnerable to relative price problems. Urban food is more expensive, causing FEI to artificially inflate urban poverty lines compared to rural ones.

$$P_{\alpha} = \frac{1}{N} \sum \left(\frac{z - y_i}{z} \right)^{\alpha}$$

Income y_i

Sensitivity Parameter α

Sample Size N

Poverty Line z

$\alpha = 0$:
Headcount Index (P_0)

Measures incidence (breadth). Fails to show how poor the poor are.

$\alpha = 1$:
Poverty Gap (P_1)

Measures depth. The minimum cost to eliminate poverty with perfect targeting.

$\alpha = 2$: Squared
Poverty Gap (P_2)

Measures severity. Weights the poorest of the poor heavily, satisfying the transfer axiom.

Master Class Challenge: Calculate the FGT Indices

The Scenario

Population:	4 citizens
Poverty Line (z):	125
Incomes (y_i):	100, 110, 150, 160

Task: Calculate P_0 (Headcount) and P_1 (Poverty Gap).

Sequential Calculations

Poor identified: 100 and 110. (2 people).

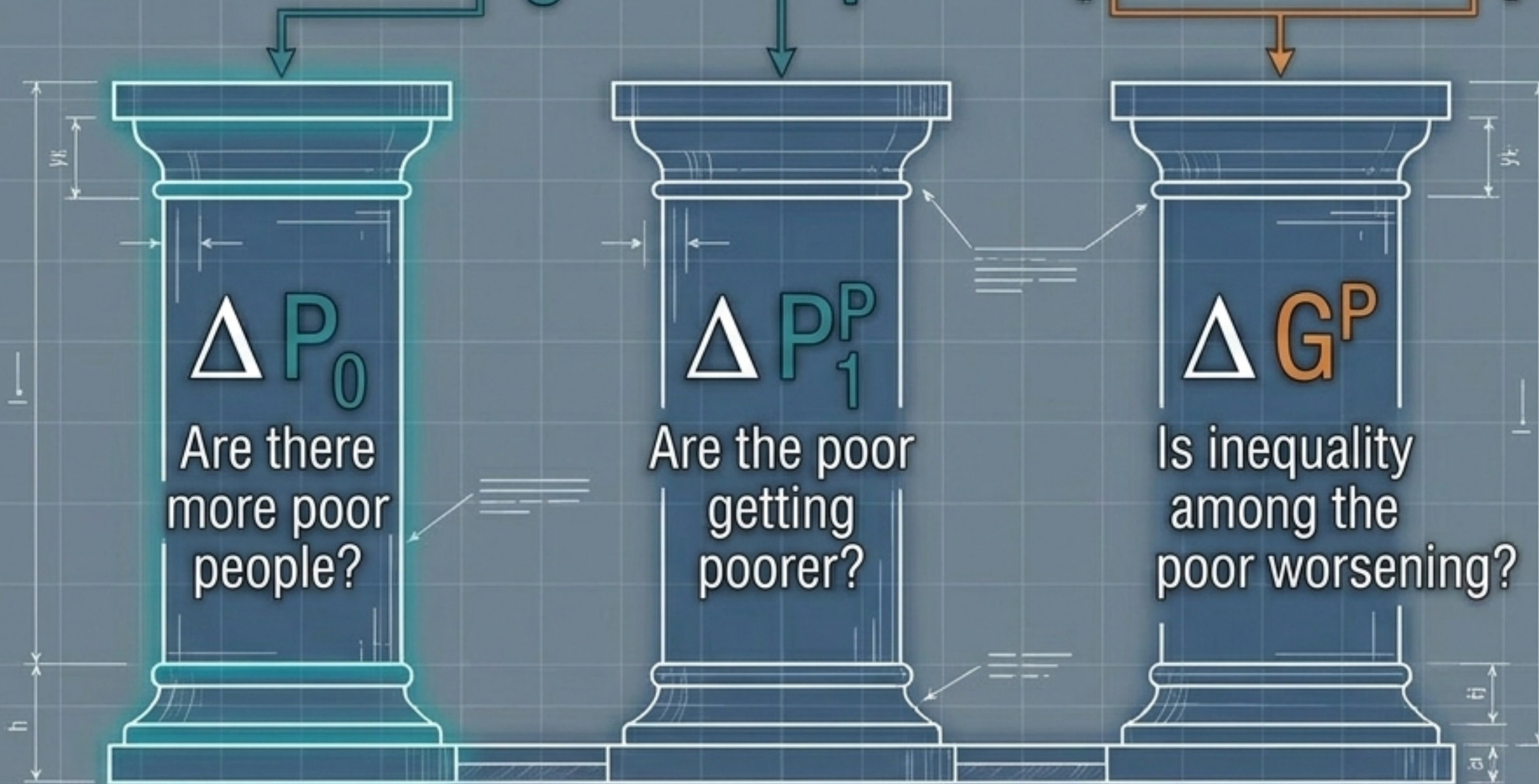
P_0 Calculation: $2 / 4 =$ **[50%]**

Gap for 100: $(125 - 100) / 125 = 0.20$

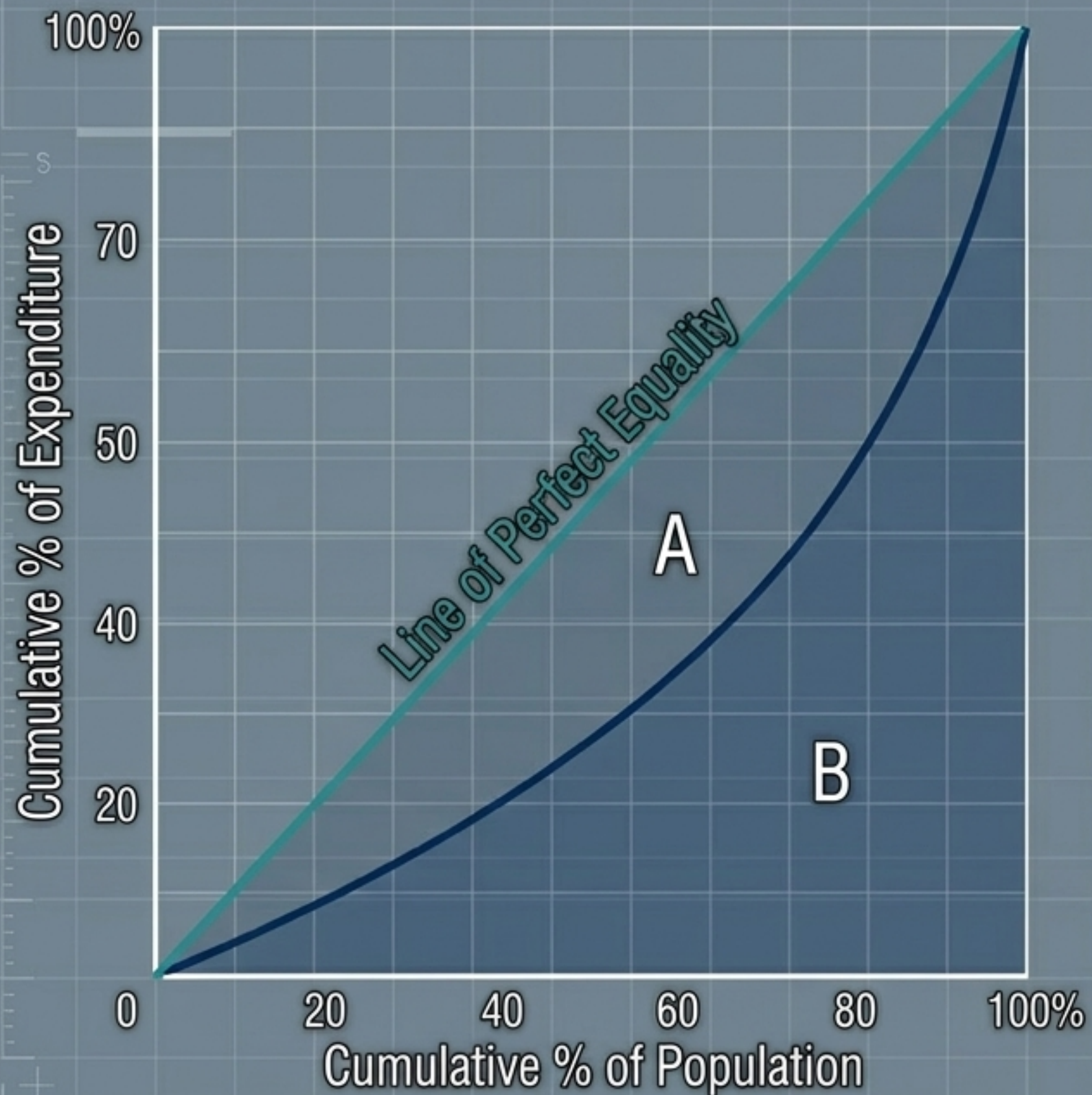
Gap for 110: $(125 - 110) / 125 = 0.12$

P_1 Calculation: $(0.20 + 0.12 + 0 + 0) / 4$
 $= 0.32 / 4 =$ **[8%]**

$$SST = P_0 \times P_1^P \times (1 + G^P)$$



By taking the log of the index, we dissolve the FGT decomposition problem and turn index changes into three direct policy questions.



The Lorenz Curve & Gini Coefficient

- Plots cumulative % of population against cumulative % of expenditure.
- Vietnam 1993 Data: Bottom 20% holds just 8.4% of expenditure; Top 20% commands 41.4%.
- Gini Coefficient = $A / (A+B)$. Ranges from 0 (perfect equality) to 1 (complete inequality).

Class Question

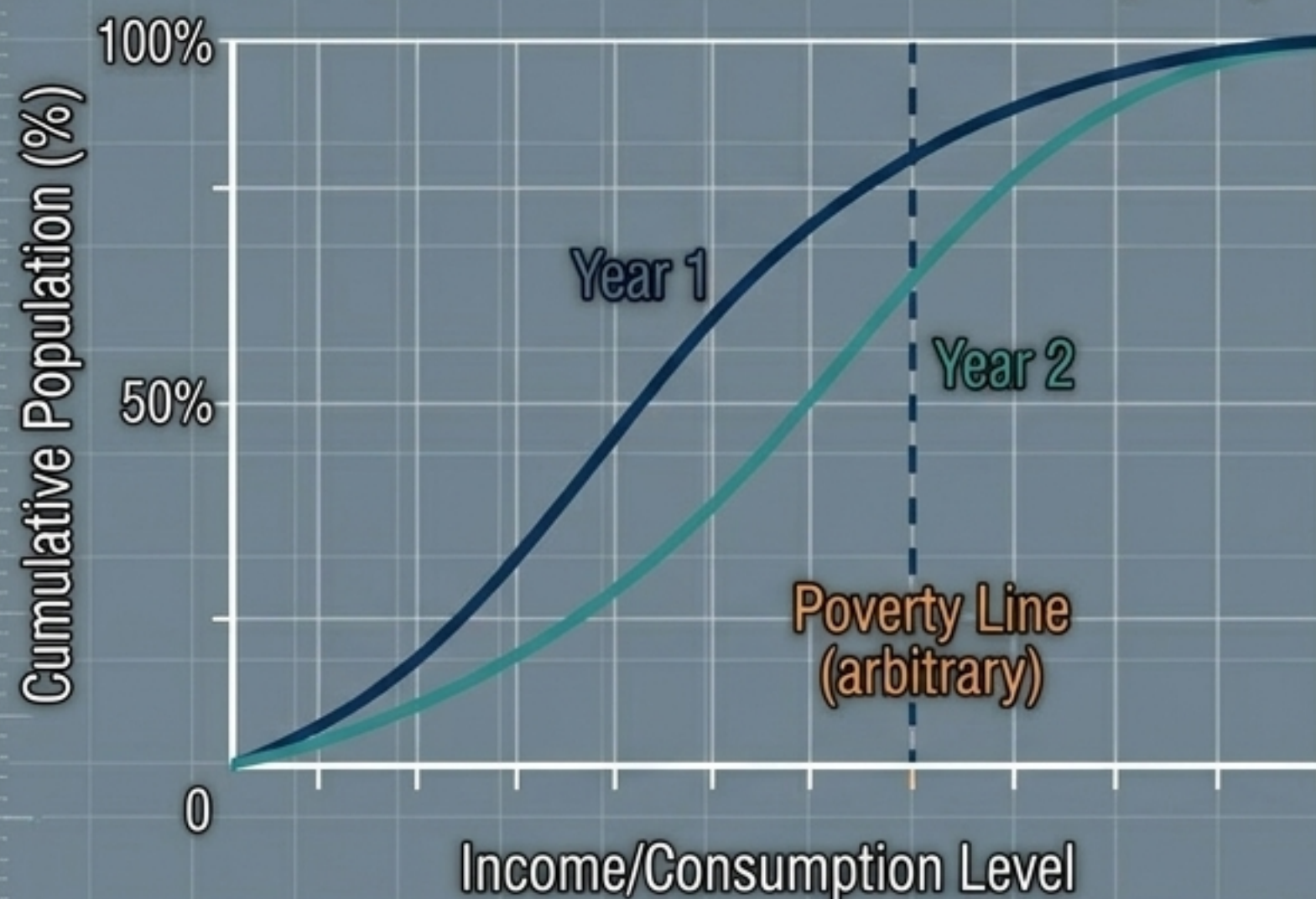
Question: If a new tax redistributes wealth from the top quintile to the bottom, how does the curve shift?

Answer: Inward, closer to the line of perfect equality.

Checking for Robustness: Stochastic Dominance

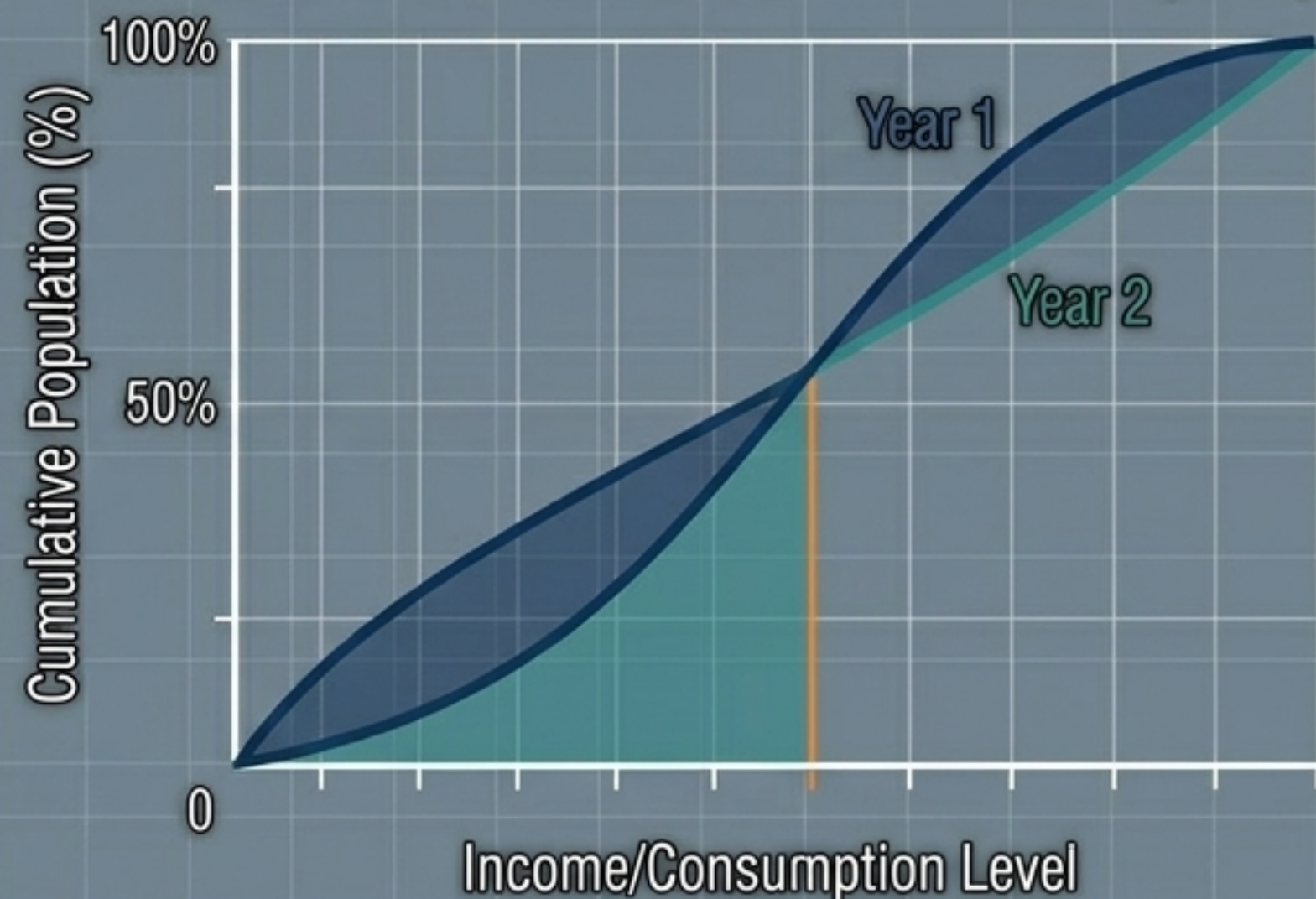
Poverty lines are arbitrary. How do we prove findings are statistically robust?

First Order Stochastic Dominance (FOD)



If the CDF of Year 2 is completely below Year 1 for all possible poverty lines, poverty has unambiguously fallen.

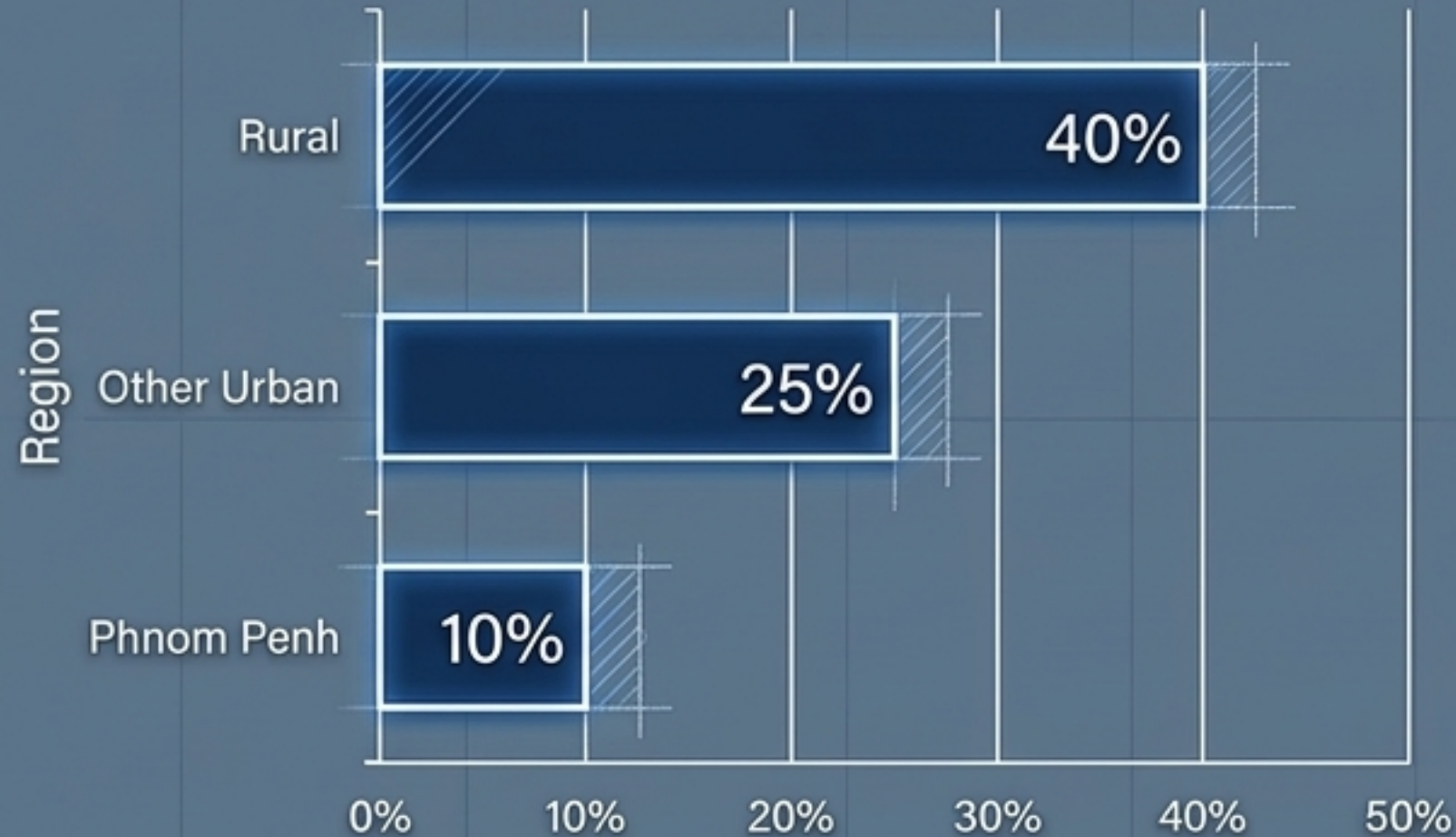
Second Order Stochastic Dominance (SOD)



If the CDF curves cross, we integrate the area under the curve. If the Year 2 deficit curve is lower, we prove poverty depth (P_1) has fallen.

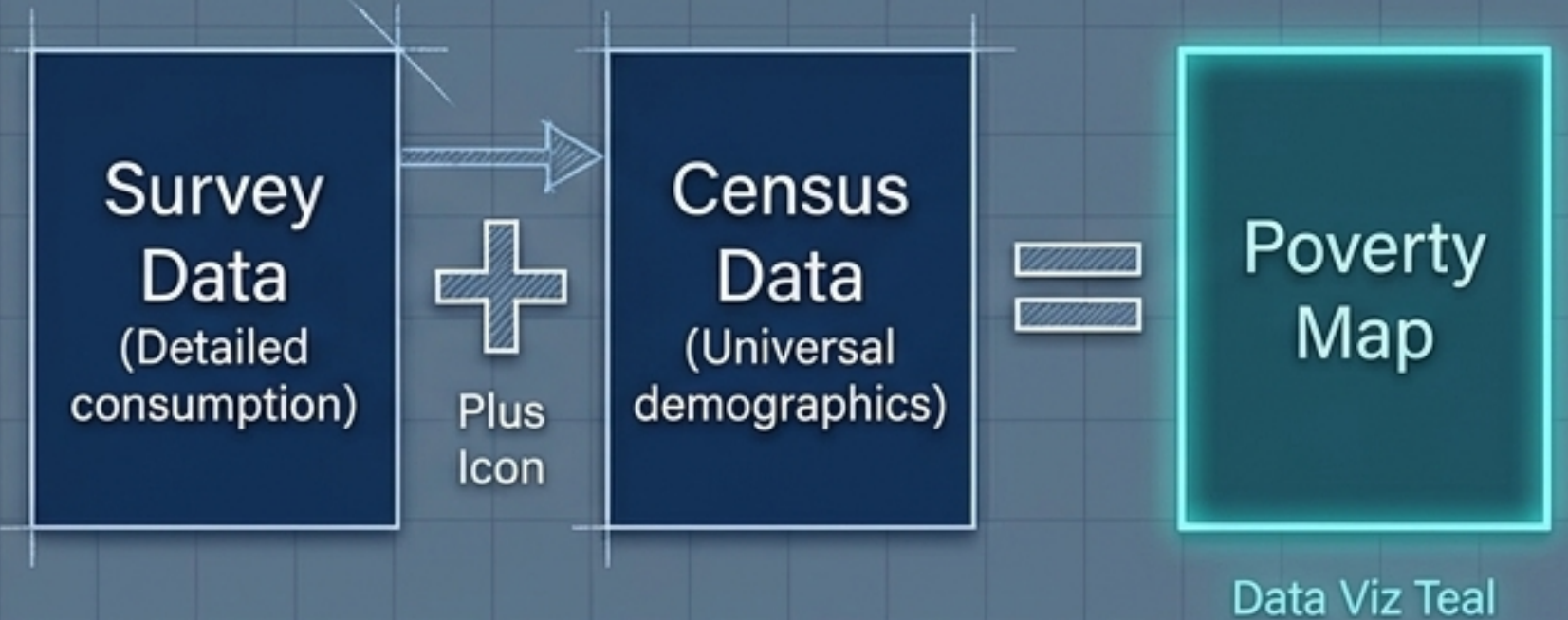
Poverty Profiling & Mapping: Spatial Analysis in Action

The Poverty Profile (Cambodia 1999)



Poverty is fundamentally **spatial**. Headcount drops dramatically as the age and education of the household head increases.

Poverty Mapping Methodology (Ecuador Case)



Surveys lack granular district-level power. By combining detailed Survey data with universal Census data via econometrics, analysts generate high-resolution maps to target block-level interventions.

$$\log(y_i/z) = \alpha_0 + \alpha_1 X_{1i} + \dots + \varepsilon_i$$

Determinants vs. Correlates

- A profile shows correlation; regression isolates **true causation** by holding other variables constant.
- **Why Log(Consumption)?** We regress on the log of per capita expenditure (rather than a binary Probit) to capture the **full** spectrum of welfare and interpret coefficients as **percentage changes**.

The Exogenous Variables (X_i)

- **Demographics:** Household size, dependency ratios.
- **Human Capital:** Years of education, individual health.
- **Physical Assets:** Land ownership, tools.
- **Community Infrastructure:** Tarred roads, clinics, regional typhoons.

Master Class Challenge: Interpreting Econometric Output

STATA/SE 8.0

```
> regress log_pcexp years_schooling rural_dummy
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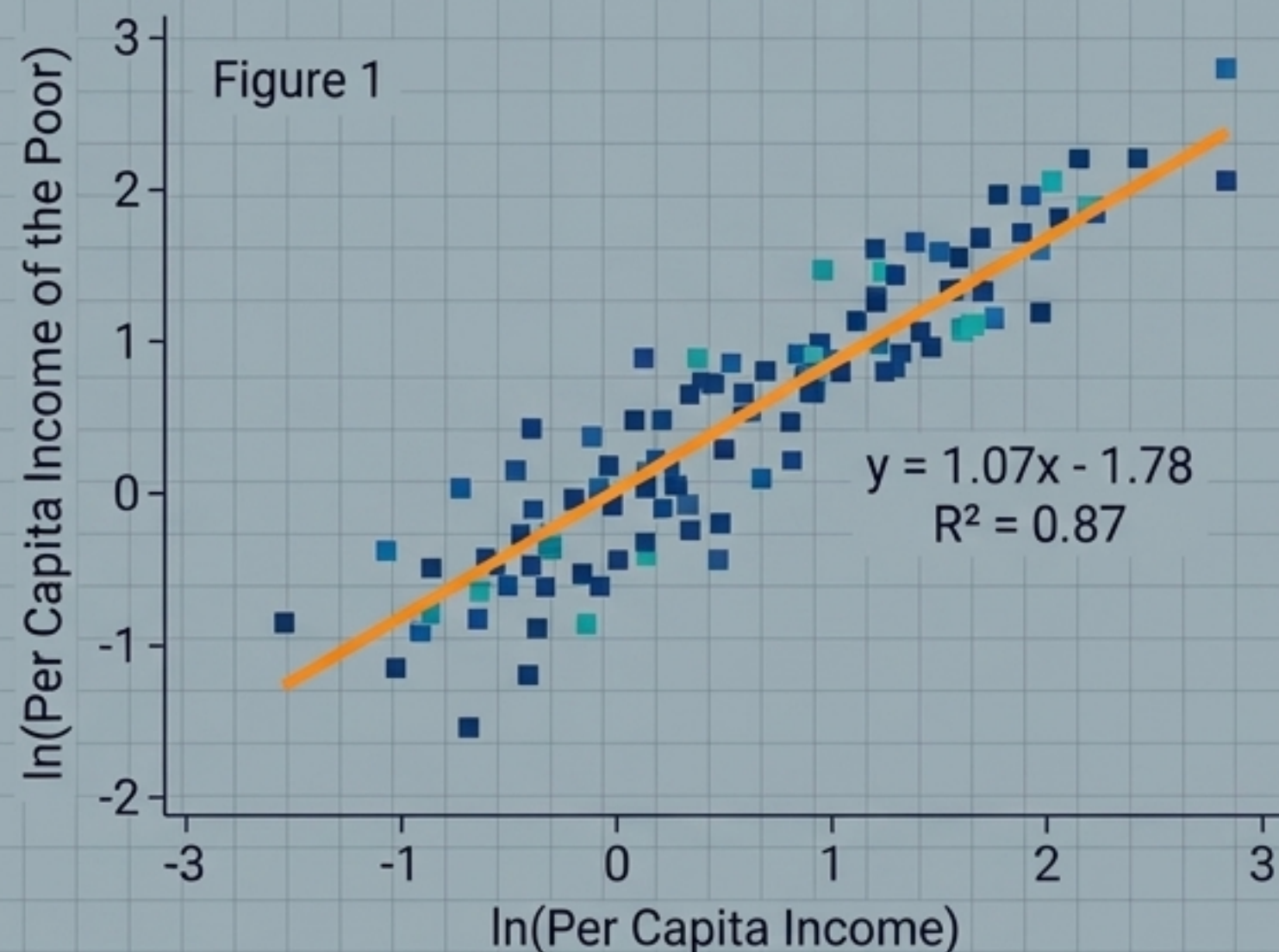
years_schooling		Coef:	0.08		P> t :	0.001
rural_dummy		Coef:	-0.15		P> t :	0.000

Master Class Challenge

How do we interpret the coefficient 0.08 for 'Years of Schooling'?

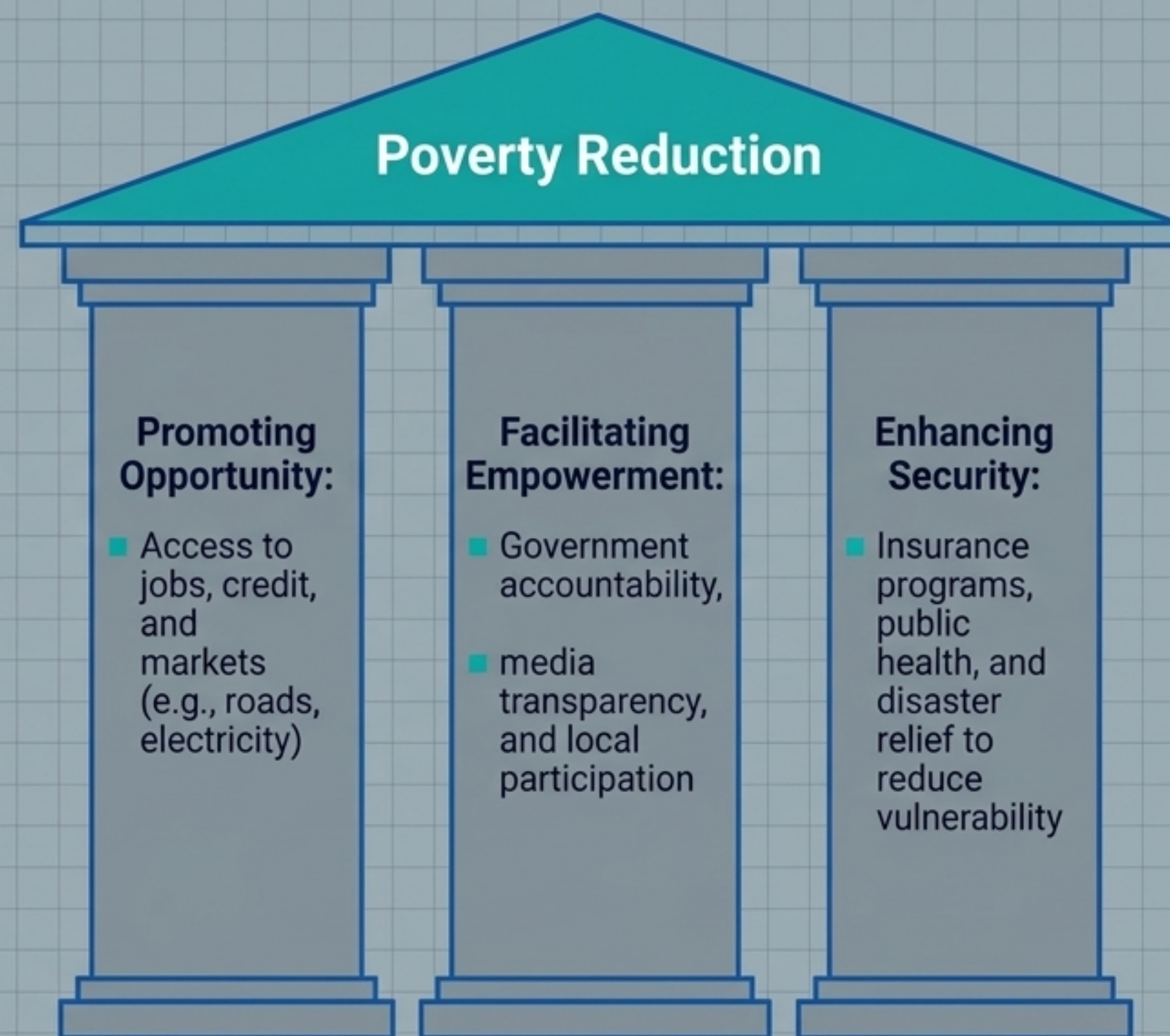
Holding all else constant, one additional year of schooling increases per capita consumption by **8%**. Because the **p-value is < 0.05**, this finding is statistically significant. ▶

The Macro Consensus



Is Growth Good for the Poor? Yes.
The elasticity of the incomes of the poorest quintile to overall per capita GDP growth is **-1.07** (Dollar & Kraay, 2000).

The 3 Pillars of Pro-Poor Policy



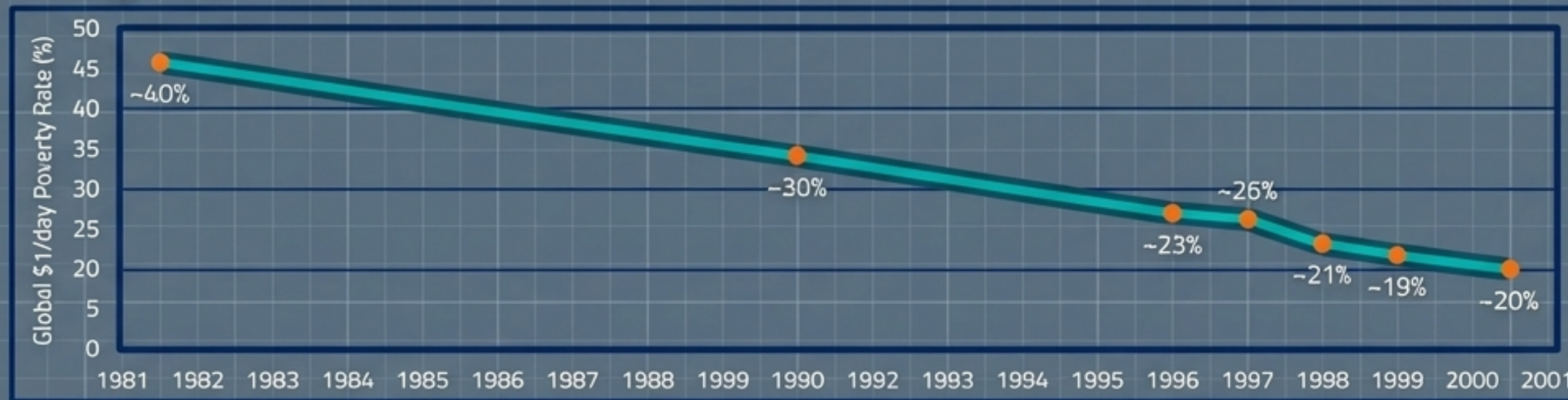
Global Stage: Measurement & Verdict

The “Dollar a Day” Standard & PPP

Purchasing Power Parity (PPP)

We cannot use market exchange rates; we must use PPP to account for cheaper non-tradable goods in developing nations to accurately track the Millennium Development Goals (MDGs).

Global Tracking



The Verdict

The Debate: Reconciling National Accounts with Household Surveys remains a statistical challenge. However, despite measurement flaws, credible and rigorous measurement remains the ultimate instrument for focusing global policy and eradicating extreme poverty.